The Palaeomagnetism of the Lupata Alkaline Volcanics

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Summary

Oriented cores have been drilled from seven sites in the alkaline volcanics of the Lupata Series, Mesozoic in age, of Mozambique. At four sites long cores (12 feet) were drilled using newly-developed sampling techniques and equipment. At the other three sites short cores were obtained with a small drill. After partial demagnetization in alternating magnetic fields, directions of magnetization were closely grouped within each site and agreement between the sites was good. These directions are considered to be those of thermo-remanent magnetization acquired when the volcanics cooled. On the assumption of an axial dipole field, the South magnetic pole parameters are

$$61°{\cdot}8\text{S.} \qquad 79°{\cdot}0\text{E.} \qquad \alpha_{95} = 4°{\cdot}1.$$

This pole position is compared with other Mesozoic poles from Africa.

1. Introduction

During August and September 1961 a collection of oriented cores was made from an exposure of the upper part of the Lupata series in the downstream end of the Lupata Gorge on the Zambesi River. Encouraging results from measurements on these cores led to further field work in this area of Mozambique during June of 1962.

The primary aim of this work was to secure reliable palaeomagnetic results from one of the few sequences of extrusive volcanic rocks post-Karroo and pre-Tertiary in age, in southern Africa. During the preceding six months a large sampling drill had been purchased and equipment for obtaining oriented core had been developed, and a secondary aim of the work here described was to carry out field trials of the new sampling equipment.

2. The geology of the Lupata Series

The Lupata Series has been described by Dixey and Campbell Smith (1929). It consists of a succession of sandstones, conglomerates, rhyolites and alkaline volcanics, the principal occurrence of which is in the Zambesi valley south-east of Tete. An excellent exposure of the series is found in the Lupata Gorge, from which the series takes its name. The sites sampled were reached via the Mungari-Tambara road and a track upstream from Tambara along the south bank of the Zambesi (Figure 1). Elsewhere access to the Lupata Series is difficult.

At the upstream end of the Lupata Gorge upper Karroo basalts are exposed (Figure 1.). Resting unconformably upon these basalts is a pebbly sandstone of

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maximum thickness 350 feet, which is regarded by Dixey and Campbell Smith as the base of the Lupata Series. Rhyolitic lavas, 260 feet thick at the entrance to the gorge, rest on the sandstone. Upon the deeply eroded surface of the rhyolite were deposited sandstones and pyroclastic rocks attaining a thickness of 450 feet

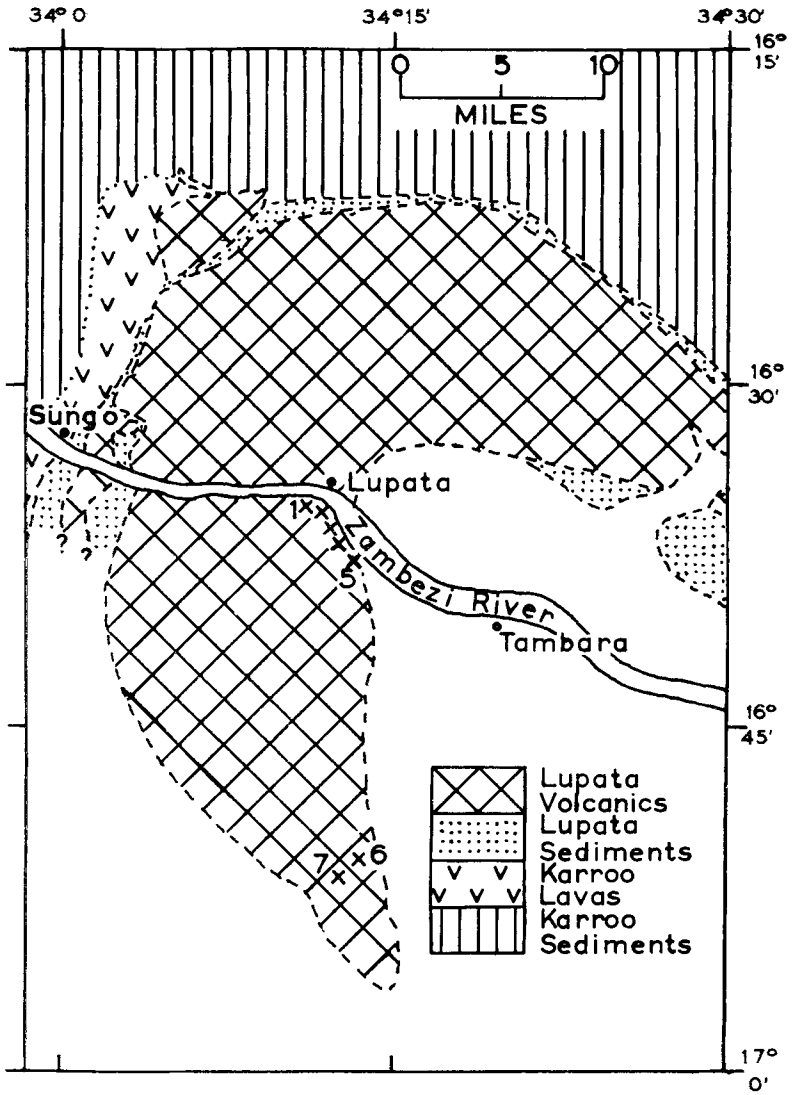


FIG. 1.—Sketch map of the Lupata Gorge area. Sampling sites are numbered 1 to 7.

in the gorge. Stratigraphically above these sediments lie alkaline lavas with associated tuffs exceeding 1100 feet in thickness, and a thick sandstone layer caps the series.

The precise age of the Lupata Series is unknown. Since its base rests upon upper Karroo basalts the series must be regarded as younger than Lower Jurassic. Pebbles from the alkaline lavas have been recorded in fossiliferous beds considered to be of Upper Cretaceous age (Dixey & Campbell Smith 1929). Provisionally, therefore, the age of the Lupata Series may be regarded as lying between Lower Jurassic and Upper Cretaceous. A shaft of a limb-bone of a large Theropodous

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Dinosaur has been found in beds correlated with the Lupata Series. This is consistent with the age limits already suggested. It may be noted that there are two unconformities between the Karroo basalts and the alkaline lavas of the Lupata Series.

The sites shown in Figure 1 are in the alkaline lavas. At site No. 8 red siltstone blocks were taken from a cutting on the main Changara-Tete road from a succession correlated with the Lupata Series. This site is about 50 miles west-north-west of the sites shown in Figure 1.

It should be noted that Figure 1, which is modified from Dixey & Campbell Smith (1929), is a reconnaissance sketch map and is not accurate in detail.

3. Sampling techniques

In African conditions, where unweathered rock exposures are rare, drilling offers a method of obtaining good material for palaeomagnetic investigation, if the cores can be oriented. A small sampling drill has been described by Graham & Keiller (1960). A drill of this type has been in use here for some years, and was used on the second visit to the Lupata Series to sample sites 4, 6 and 7. As part of a programme of development supported by the National Science Foundation of the United States of America, a prospecting drill was purchased in 1961. This drill is capable of drilling cores $1\frac{3}{8}$ inches in diameter to a depth of several hundred feet. Orientation techniques have been developed which enable cores to be oriented to a depth of twenty feet. These techniques could be applied at considerably greater depths if desired.

The rotary drill has a turret through which the drill rods pass, and which is advanced by a hand-operated rack and pinion. The turret swivels about a horizontal axis to allow holes to be drilled at various angles to the vertical. A single-cylinder four-stroke 0·38 litre petrol engine drives the drill through a three-speed gearbox. The drill and engine are permanently mounted on a trailer, with an aperture in the floor through which the drill-string* passes. In use the trailer wheels are removed and the trailer stands on three demountable tetrahedral legs. The trailer is firmly anchored to the rock outcrop by four wire ropes with turn buckles secured to expanding bolts in holes previously drilled with the Graham–Keiller drill. The drill is supplied with water from a small independent pumping unit. The drill rods terminate in a brass core-barrel carrying a bronze diamond crown.

A given orienting mark can be extended down the core as long as any breaks in the core can be uniquely fitted. When a break occurs which cannot be fitted, the drill-string is withdrawn and the core-spring, which normally enables the core to be broken and lifted, is removed from the crown. A three inch length of core is now drilled and left standing in the hole. The striker unit, marked A in Figure 2, has a diamond mounted on a steel leaf-spring which is held retracted in the striker wall when the trigger B is in the extended position shown. The striker unit is attached to a duralumin tube T which carries a scratch along its whole length, in register with the diamond. The tube T is passed through the turret and the striker is lowered nearly to the bottom of the hole. The orienting instrument marked C is clamped to the tube above the turret, in alignment with the scratch on the tube, as shown in Figure 2. The chuck is now rotated until the transverse bubble at D is level, with the scratch on the lower side of the tube. The front part of the orienter is rotated about the axis EF to level the longitudinal bubble at D.

* The drill-string is the whole assembly of crown, core-barrel and drill rods.

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The inclination of the core is read from the quadrant *G*. It is necessary to determine the azimuth of the vertical plane containing the axis of the core. The vertical line on plate *H* lies along the axis of the graduated half-cylinder *J*. When the pointer *K* is set so that its shadow falls on this line, the required azimuth can be

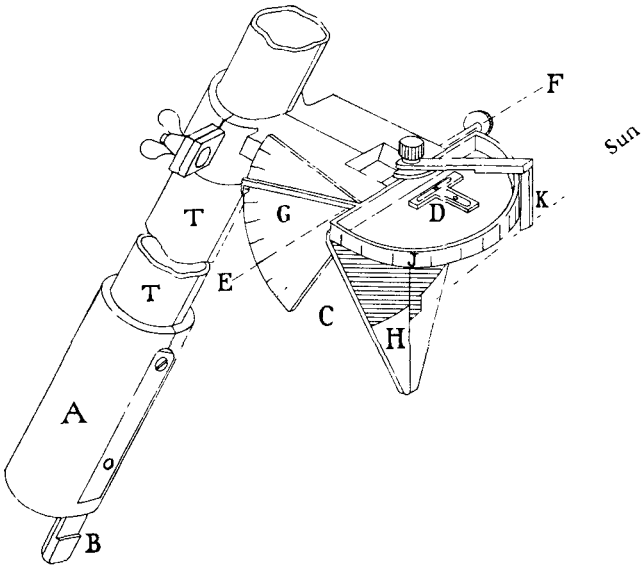


FIG. 2.—Instruments for orienting rock core. The reference letters refer to the text.

found from the Sun's azimuth at the time of the observation, and a correction read from the half-cylinder *J* at the pointer *K*. The turret is now advanced until trigger *B* releases the spring-mounted diamond. Finally the turret is raised, to produce a longitudinal scratch on the core. Drilling is continued.

If the Sun is unsuitably placed for direct orientation, a theodolite is used to transfer the sun's azimuth, or any convenient known azimuth, to the orienter. With the theodolite sighted upon the vertical line on plate *H*, the half-cylinder *J* is read through the theodolite. The orientation procedure used with the small drill differs in detail from that described, but is identical in principle. In both cases the orientation error is unlikely to exceed one degree.

The time required to set up the large drill and obtain ten feet of oriented core is 1½ to 2 days. For comparison, ten short cores can be drilled with the small drill in about six hours. Both drills with all auxiliary equipment and camping gear can be carried in a long wheel-base Land Rover and trailer, and constitute a powerful sampling unit.

4. Field work

At each of sites 1, 2, 3 and 5 a core was drilled with the large drill during the 1961 visit. These cores ranged from six to twelve feet in length. From each long core nine specimens were cut distributed along the whole length, and the combination of the directions of magnetization of these specimens is regarded as a site mean direction of magnetization. At sites 4, 6 and 7 from seven to ten short cores were drilled. Two specimens were measured from each short core, and the mean core directions of magnetization were combined to give a site mean.

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Site 1 is 11 miles upstream from Tambara. From Site 1 the distances of sites 2 to 5 are:

Site	2	3	4	5
Distance	0·4	1·7	2·6	5·8 miles

Sites 1 and 2 are known to be in successive lava flows because the contact has been located in the field. The rocks differ widely in petrologic character at these two sites (Table 1). It is possible that site 3 is in the same flow as site 2, though the distance between the sites makes this improbable. Site 4 is known to be in a stratigraphically higher flow than site 3, because pyroclastic rocks intervene. Site 5 is separated from Site 4 by a large distance and the rocks are petrologically distinct.

Sites 6 and 7 are located about seventeen miles south of the other sites. They are about 2 miles apart, and are probably situated in different flows. It is unlikely that these sites represent the same stratigraphic horizon as the river-bank sites.

Table 1 gives the best information that could be obtained in the field regarding the attitude of the rocks. Some uncertainty exists in the strike and dip determinations, and this may be the major source of error in the final pole position. Such error is unlikely to exceed five degrees at any site.

Table 1

Site	*Strike*	*Dip*	*Rock type*
1			Quartz trachyte
2			Kenyite
3	35°	10° SE.	Kenyite
4			Kenyite
5			Altered analcite phonolite
6	320°	5° NE.	Fine-grained phonolite
7			Fine-grained phonolite
8	346°	5° W.	Red siltstone

5. Palaeomagnetic results

Directions of magnetization were measured using a spinner magnetometer to be described elsewhere. The directions of natural remanent magnetization are given in Table 2. At four sites the directions of N.R.M. are widely scattered. At three sites the directions of N.R.M. are well grouped, Fisher's K being 188 for site 3. The mean directions from these three sites do not, however, agree. At all sites the effects of lightning currents would account for the observed directions of N.R.M.

All specimens have been treated in alternating magnetic fields using the apparatus described by McElhinny & Gough (1963). Standard empirical procedures were followed to determine the best demagnetizing fields (As & Zijderveld 1958, Creer 1959, Irving, Stott & Ward 1961). All specimens at all the sites reached stable directions of magnetization with increasing field.

The parameters describing the magnetizations after alternating field treatment are given in Table 2, and the directions of magnetization are shown in Figures 3 and 4. These directions have not been corrected for any assumed movement of the rocks after magnetization. In each case the value of K (Fisher 1953) shows a marked increase after partial demagnetization, and the site mean directions agree closely.

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Figure 3 shows the directions of magnetization, before and after alternating field treatment, at sites 1 and 2. At site 1 the distribution of directions of N.R.M.

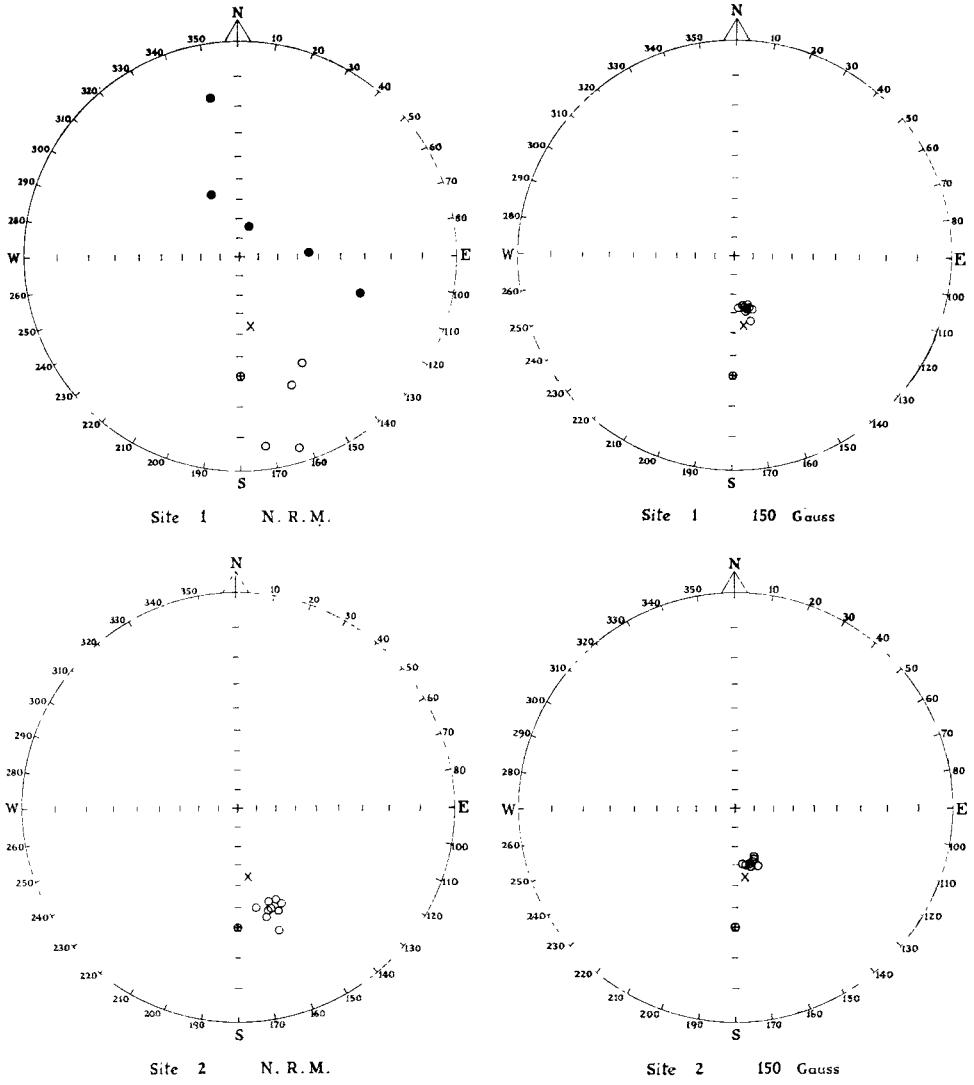


FIG. 3.—Directions of remanent magnetization at sites 1 and 2, before and after treatment in alternating fields; Lower hemisphere stereographic projection.

○ South-seeking pole } downward
● North-seeking pole }
× Direction of present geomagnetic field
⊕ Direction of axial dipole field.

strongly suggests remagnetization by a close lightning flash, the displacements from the direction of thermo-remanent magnetization increasing steadily up the core (Graham 1961, Cox 1961). The stereograms for site 2 are instructive. The N.R.M. directions form a close group which moves bodily on partial demagnetization in 150 gauss. The value of K increases from 26·7 to 941.

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Table 2

		Natural remanent magnetization				*Magnetization after treatment in alternating fields*							
Site	N	D	I	R	*K*	D	I	R	*K*	α_{95}	B	D_c	I_c
1	9	349°·7	+21°·9	6·0199	2·68	346°·7	−60°·9	8·9880	667	2°·0	500	337°·6	−53°·5
2	9	343 ·3	−37 ·1	8·9700	26·7	346 ·2	−60 ·3	8·9916	941	1 ·6	150	337 ·0	−52 ·9
3	9	348 ·6	−62 ·9	8·9574	188	338 ·2	−64 ·6	8·9853	544	2 ·2	150	329 ·5	−56 ·1
4	7	117 ·1	−23 ·7	0·7797	0·965	332 ·2	−59 ·5	6·9563	137	5 ·2	400	326 ·8	−50 ·3
5	9	96 ·5	−20 ·6	8·7884	37·8	356 ·3	−64 ·7	8·9956	1800	1 ·2	150	343 ·9	−58 ·0
6	8	310 ·8	−21 ·3	3·7068	1·63	344 ·8	−54 ·9	7·9685	222	3 ·70	500	339 ·0	−56 ·7
7	10	339 ·6	+38 ·0	8·5574	6·24	344 ·8	−50 ·2	9·8176	49·3	6 ·9	300	339 ·0	−52 ·6
8	5	2 ·2	−52 ·7	4·5227	8·38	353 ·4	−46 ·9	4·9750	160	6 ·1	100	358 ·8	−46 ·0

N = number of specimens measured from a long core at sites 1, 2, 3 and 5
= number of short cores at sites 4, 6 and 7.
= number of blocks collected at site 8.

D = declination
I = inclination
D_e = declination
I_c = inclination
} corrected to restore strata to the horizontal.

R = Resultant of N unit vectors.

$$K = \frac{N-1}{N-R}$$

α_{95} = semi-angle of cone of 95 per cent confidence for mean direction.
B = maximum peak alternating field in gauss.

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At site 7 specimens from two short cores gave directions of magnetization which moved, on demagnetization in increasing fields, towards the group given by the other cores. Both reached stable directions outside the group, and it is possible that they have been strongly affected by lightning. This supposition cannot be proved and all cores have been used in deriving the site mean direction. The directions of magnetization are closely grouped at each site, as will be seen in Table 2 and Figures 3 and 4. This is believed to result from accurate orientation procedures applied to rocks of favourable magnetic properties.

It will be observed that the directions of magnetization, after treatment in alternating fields, lie close to the direction of the present geomagnetic field, though much less close to the axial dipole field direction. These magnetizations are very little affected in magnitude or direction when treated in alternating fields which are increased progressively from the values indicated in Figures 3 and 4, to 800 gauss. They are, therefore, stable magnetizations of high coercivity. The possibility exists that the rocks have been chemically magnetized in recent times in the present field. This is improbable, however. The data are derived from lava flows of various petrologic types (Table 1), and in four cases from nearly vertical cores six feet or more long. The hypothesis of recent chemical magnetization would require that at the seven sites, over the depths sampled and irrespective of petrologic variation, magnetic oxides had been formed within two or three hundred years, i.e. while the field had its present large displacement from the dipole field direction. Chemical magnetization at more probable, varying rates, would presumably produce directions of magnetization scattered on account of secular variation, but not concentrated near the present field direction. For these reasons it is believed that the directions of magnetization, after treatment in alternating fields, are those of thermoremanent magnetization at the time of extrusion of the lavas. This view receives some support from the fold test reported in Section 6 below.

The circles of 95 per cent confidence for the mean directions of magnetization at the sites after treatment in alternating fields, are shown in Figure 5. The directions there shown have been corrected to restore the strata at each site to the horizontal.

6. Discussion

It can be seen from Figure 5 that the red siltstone sampled at site 8 gives a direction of magnetization significantly different from those given by the volcanics. The correlation of this siltstone with the Lupata Series is not secure, and sites 1 to 7 only have been used in calculating a pole position.

Sites 1 to 5 cover a minimum thickness of 2 500 feet of volcanics, even if the dip is only half that shown in Table 1. Since sites 6 and 7 are some 17 miles from the river-bank sites, in rock types not seen by the authors along the Zambesi River, it is reasonable to presume that these sites are either higher or lower stratigraphically than the sites sampled along the Zambesi. As six or seven flows have been sampled over a considerable thickness of volcanics, it is probable that a mean pole position calculated from the seven sites will approximate to the geocentric axial dipole position freed from secular variation. If this is correct, the close grouping of the site mean directions of magnetization suggests that the amount of secular variation was rather small at the time of cooling of the Lupata volcanics. As will be seen from Figure 5, there are nevertheless significant differences between site mean directions, because of the low within-site dispersions. Using pole positions computed for individual sites, the standard deviation is 5 degrees,

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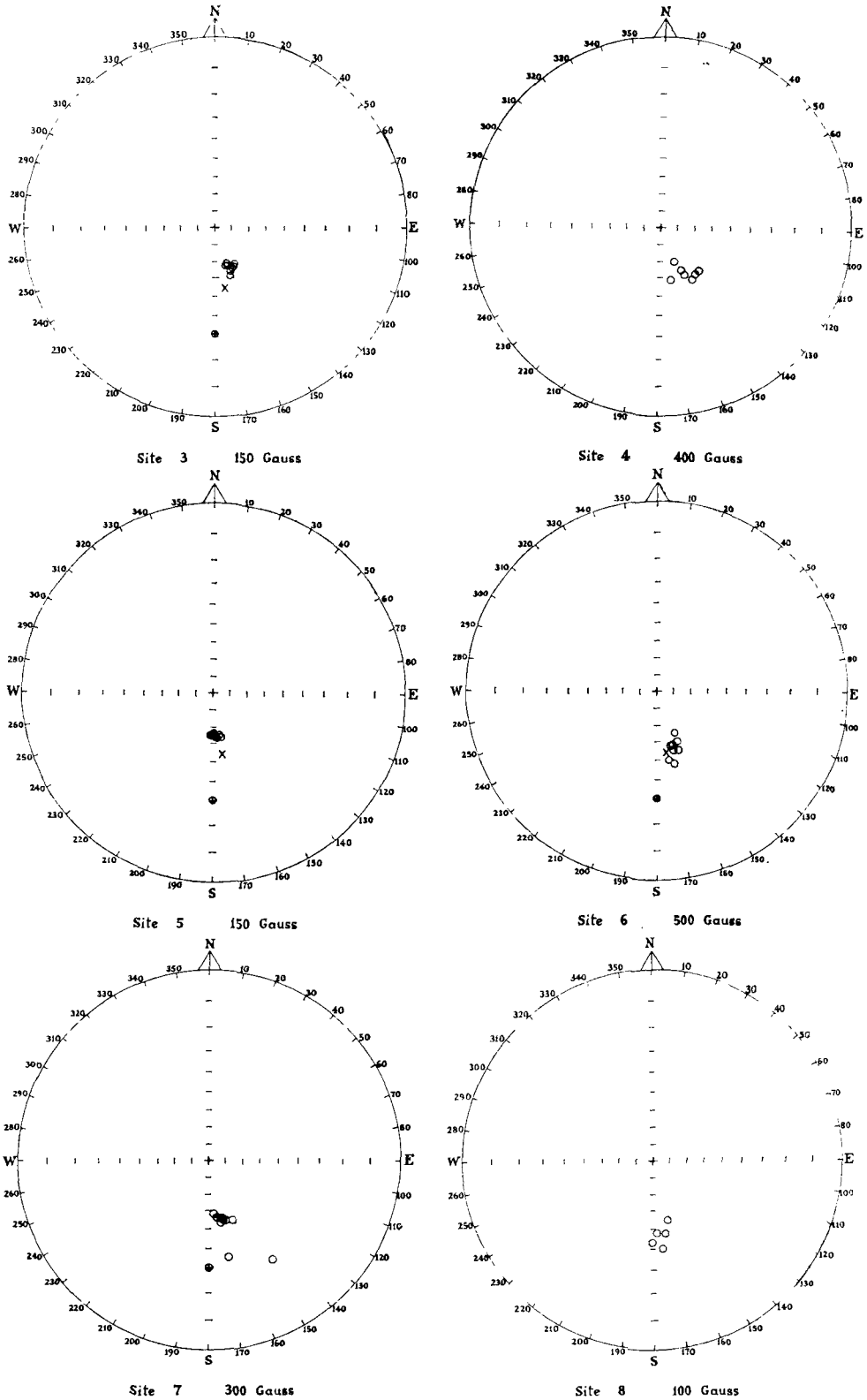


FIG. 4.—Directions or remanent magnetization after treatment in alternating magnetic fields at sites 3 to 8. Lower hemisphere stereographic projection.

○ South-seeking pole downward
× Direction of present geomagnetic field
⊕ Direction of axial dipole field.

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which is about one-third of the standard deviation found by Vestine & others (1947) for the present-day variations of the geomagnetic field from an axial dipole field.

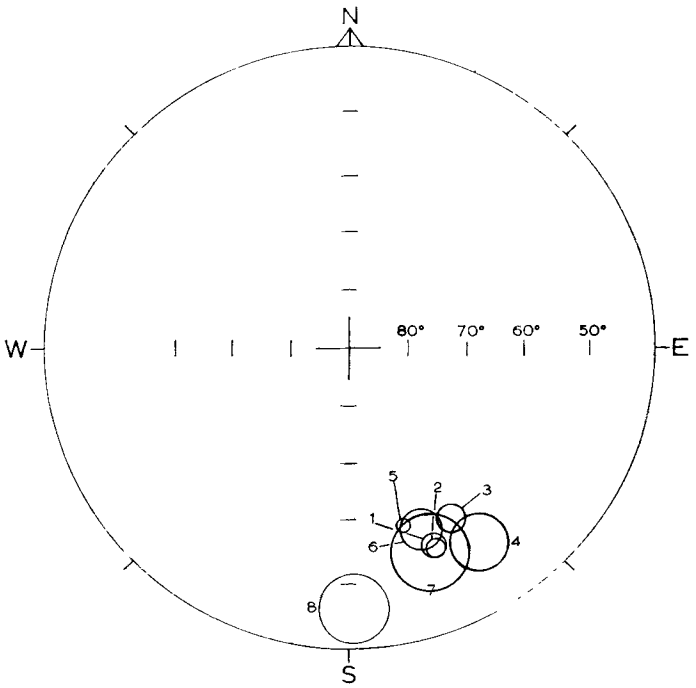


FIG. 5.—Circles of 95 per cent confidence for site mean directions of magnetization, corrected to restore the strata to the horizontal.

As is shown by Table 1, the Lupata volcanics have only small dips at the sampling sites. Nevertheless, as a result of the low within-site dispersion the effect of applying corrections for geological dip is appreciable. The mean declination, inclination and Fisher's K parameter, for the seven mean directions after alternating field treatment, are:

	D	I	K
Without dip correction	344°·2	−59°·5	162
With dip correction	335°·8	−54°·4	339

The increase in K on restoring the strata to the horizontal suggests that the lavas were more nearly horizontal when magnetized than they are now, and that the thermoremanent magnetization has been stable since the displacements occurred (Graham 1949).

The mean magnetic pole position, assuming a geocentric axial dipole field, has been calculated in two ways. From the mean direction of magnetization of the seven sites, the South magnetic pole position is

$$61°{\cdot}8\text{S.} \quad 79°{\cdot}5\text{E.} \quad \delta p^* = 3°{\cdot}2 \; \delta m^* = 4°{\cdot}5$$

From pole positions computed from individual site means, the pole position is:

$$61°{\cdot}8\text{S.} \quad 79°{\cdot}0\text{E.} \quad \alpha_{95} = 4°{\cdot}1.$$

* δp is the semi-axis of the oval of 95 per cent confidence for the pole position, along the great circle through the site; δm is the other semi-axis.

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It has been suggested by McElhinny & Gough (1963) that the second method of calculating the mean pole position is preferable where the principal cause of variation of site mean direction is secular variation. As this is believed to be true in the present study, the circle of confidence given by the second method is shown in Figure 6.

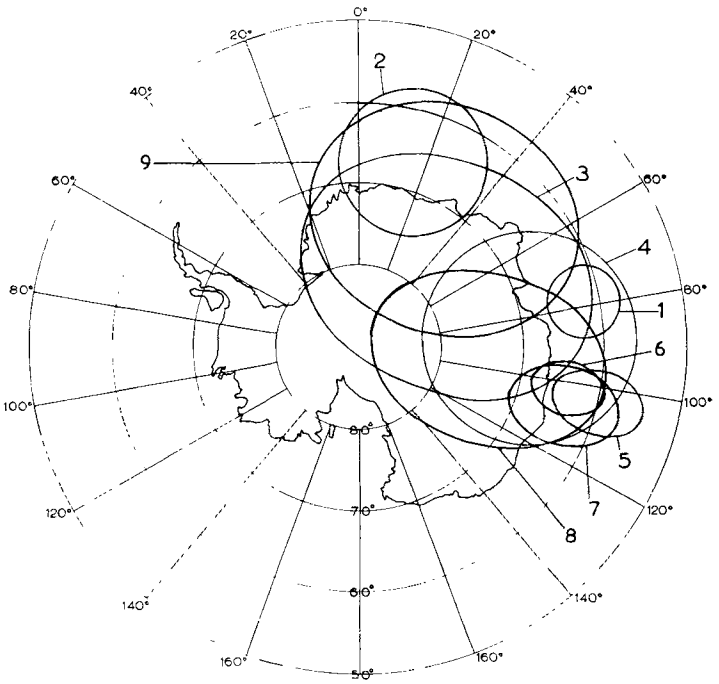


FIG. 6.—Pole positions from the Mesozoic of Africa
(95 per cent confidence limits)

1. Lupata volcanics.
2. Cretaceous lavas and dykes of Madagascar (Roche & Cattala 1959, pole calculated by Cox & Doell 1960).
3. Karroo dolerites, surface samples (Graham & Hales 1957, pole calculated by Irving, 1959).
4. As 3, underground samples.
5. Karroo basalts (Nairn 1960, pole calculated by Irving 1959).
6, 7, 8, 9. Karroo basalts of Basutoland (van Zijl, Graham & Hales, 1962).
6. Normally magnetized flows, Sani Pass.
7. Normally magnetized flows, Masery district.
8. Reversely magnetized flows, Sani Pass.
9. Reversely magnetized flows, Maseru district.

The Lupata volcanics give a pole position which is in substantial agreement with those given by the Karroo dolerites and basalts (Figure 6). These latter are generally regarded as Lower Jurassic in age. However the Cretaceous pole position from Madagascar differs considerably from the Lupata pole. The difference is not consistent with the hypothesis of simple separation of Madagascar from Africa without rotation. It should be noted that only pole positions 1, 6, 7, 8 and 9 (Figure 6) are derived from measurements on rock specimens previously treated in alternating fields. It is possible that the uncertainties in the other pole positions would be reduced if the rocks were subjected to partial demagnetization in alternating fields. While the age of the Lupata Series is subject to some uncertainty,

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it is certainly considerably younger than the Karroo basalts. It appears that the pole may not have moved greatly relative to Africa during the Jurassic and Cretaceous Periods.

Acknowledgments

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